Create and Read Documents

In MongoDB, data is stored in **documents** (which are JSON-like objects) inside **collections**. You can perform **Create** and **Read** operations using simple methods.

This section will cover:

- `insertOne()`, `insertMany()`
- `find()`, `findOne()`
- Basic filters and projections

Note: All code examples in this section are written for **MongoDB Compass** (MongoDB Shell syntax). You can run these directly in the MongoDB Compass shell or mongosh.

Setting Up Sample Data

Before we start, let's create a school database with students and teachers. Run this in MongoDB Compass:

```
// Switch to school database
use school

// Insert sample teachers
db.teachers.insertMany([
  {
    _id: ObjectId("507f1f77bcf86cd799439011"),
    name: 'Dr. Kumar',
    subject: 'MongoDB',
    experience: 5
  },
  {
    _id: ObjectId("507f1f77bcf86cd799439012"),
```

```
    name: 'Prof. Sharma',
    subject: 'Node.js',
    experience: 8
  },
  {
    _id: ObjectId("507f1f77bcf86cd799439013"),
    name: 'Ms. Patel',
    subject: 'Express',
    experience: 3
  }
])

// Insert sample students with teacher references
db.students.insertMany([
  {
    name: 'Ali',
    age: 22,
    course: 'MongoDB',
    enrolled: true,
    teacherId: ObjectId("507f1f77bcf86cd799439011"),
    grades: [85, 90, 88]
  },
  {
    name: 'Sara',
    age: 20,
    course: 'Node.js',
    enrolled: true,
    teacherId: ObjectId("507f1f77bcf86cd799439012"),
    grades: [92, 88, 95]
  },
  {
    name: 'Ahmed',
    age: 24,
    course: 'Express',
    enrolled: true,
    teacherId: ObjectId("507f1f77bcf86cd799439013"),
    grades: [78, 82, 85]
  },
  {
    name: 'Fatima',
```

```
    age: 21,
    course: 'MongoDB',
    enrolled: true,
    teacherId: ObjectId("507f1f77bcf86cd799439011"),
    grades: [95, 93, 97]
  },
  {
    name: 'Ravi',
    age: 23,
    course: 'Node.js',
    enrolled: false,
    teacherId: ObjectId("507f1f77bcf86cd799439012"),
    grades: [70, 75, 72]
  }
])
```

Inserting Documents

insertOne

Use this to insert a single document into a collection.

```
db.students.insertOne({
  name: 'Priya',
  age: 19,
  course: 'MongoDB',
  enrolled: true,
  teacherId: ObjectId("507f1f77bcf86cd799439011"),
  grades: [88, 91, 89]
})
```

insertMany

Use this to insert multiple documents at once.

```
db.students.insertMany([
  {
    name: 'Kabir',
    age: 20,
    course: 'Node.js',
    enrolled: true,
    teacherId: ObjectId("507f1f77bcf86cd799439012"),
    grades: [84, 87, 86]
  },
  {
    name: 'Zara',
    age: 22,
    course: 'Express',
    enrolled: true,
    teacherId: ObjectId("507f1f77bcf86cd799439013"),
    grades: [90, 92, 94]
  }
])
```

Reading Documents

findOne

Returns the first document that matches the filter.

```
db.students.findOne({ name: 'Ali' })
```

find

Returns all matching documents. In MongoDB Compass, results are automatically displayed.

```
db.students.find({ course: 'MongoDB' })
```

To get all students:

```
db.students.find({})
```

Projections

Use projections to include or exclude specific fields.

```
// Include only name and age (exclude _id)
db.students.find({}, { _id: 0, name: 1, age: 1 })
```

```
// Include name, course, and grades
db.students.find({}, { name: 1, course: 1, grades: 1 })
```

Other Options

Limiting Results

```
db.students.find().limit(3)
```

Sorting Results

```
// Sort by age (descending)
db.students.find().sort({ age: -1 })
```

Combining Operations

```
// Find MongoDB students, show only name and grades, sorted by age
db.students.find(
  { course: 'MongoDB' },
```

```
  { name: 1, grades: 1, _id: 0 }
).sort({ age: 1 })
```

Working with Teachers Collection

```
// Find all teachers
db.teachers.find()

// Find teacher by subject
db.teachers.findOne({ subject: 'Node.js' })

// Find experienced teachers (more than 5 years)
db.teachers.find({ experience: { $gt: 5 } })
```

Summary

- Use `insertOne()` and `insertMany()` to add data
- Use `findOne()` and `find()` to read data
- Use projections to control which fields are returned
- Use `.sort()` and `.limit()` to customize results
- All code runs directly in MongoDB Compass shell